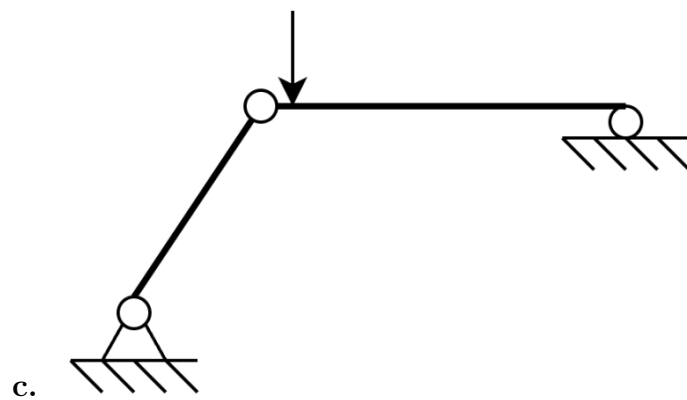
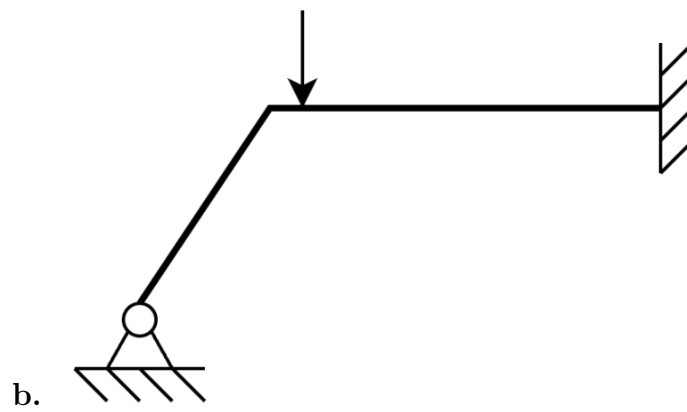
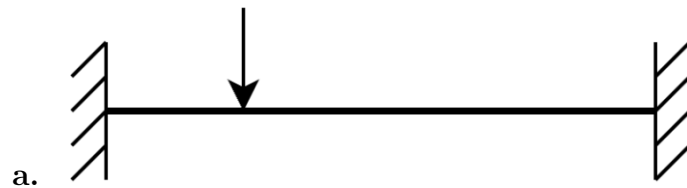


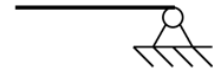
CE 430, Homework 1

Q1

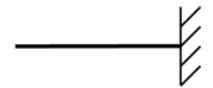
For the structures shown below, specify if they are determinate, indeterminate (and to what degree), or unstable. (Note: open circle denotes a pin.)



Roller support



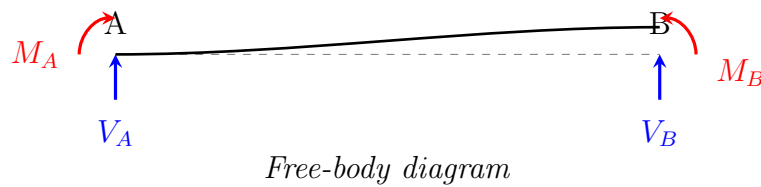
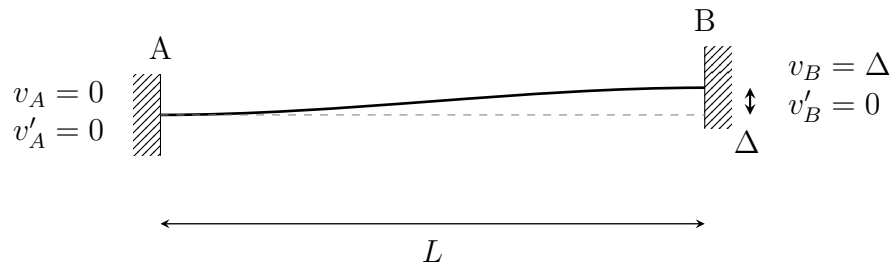
Pin support



Fixed end

Q2

Using the same approach as in class, derive the moment function $M(x)$ and the shear function $V(x)$ for the relative displacement in a flexural member, as shown, where x is the horizontal coordinate measured from end A : $x = 0$ at A and $x = L$ at B .



Q3

Using the same approach as in class, derive the moment function $M(x)$ and the shear function $V(x)$ for the load case below, where $x = 0$ at A and $x = L$ at B . Then **verify the fixed-end moments** by evaluating your moment function at the two ends: check that $M_A = M(0) = \frac{wL^2}{30}$ and $M_B = M(L) = \frac{wL^2}{20}$.

